

Program overview

19-May-2025 11:41

Year 2024/2025
Organization Civil Engineering and Geosciences
Education Schakelopleiding Civil Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Schakel Programma MSc CIE: Hbo Diploma Civiele Techniek (2024)							
Civil Engineering Allround. Compulsory courses for all students, adding up to 35 EC.							
CT1730HBO	Introduction to Geotechnical Engineering	3					
CT2023	Programmeren in Python	3					
CTB1210	Dynamica & Modelvorming	5					
CTB2300	Dynamics of Systems	3					
CTB2400	Numerical Mathematics	3					
IFEEMCS010500	Probability and Statistics	3					
IFEEMCS012100	Calculus for Engineering, part 1	3					
IFEEMCS012200	Calculus for Engineering, part 2	3					
IFEEMCS012300	Calculus for Engineering, part 3	3					
WI1807TH1-21	Linear Algebra	3					
WI1909TH	Differential Equations	3					
Civil Engineering Fundamentals. Students choose one course out of the following two Civil Engineering Fundamentals courses.							
CT1000 S	Structural Mechanics	6					
CTB2110	Fluid Mechanics	5					
Civil Engineering Design. Students choose one course out of the following four Civil Engineering Design courses.							
CTB3340-15	Building Structures 1	4					
CTB3355	Hydraulic Structures	4					
CTB3370-18	Geometrical Design of Roads and Railways	4					
CTB3420	Integral Design of Infrastructure	4					

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Schakelopleiding Civil Engineering

Schakel Programma MSc CIE: Hbo Diploma Civiele Techniek (2024)

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Schakelopleiding Civil Engineering

Civil Engineering Allround. Compulsory courses for all students, adding up to 35 EC.

CT1730HBO	Introduction to Geotechnical Engineering	3
Responsible Instructor	Prof.dr. M.A. Hicks	
Instructor	Ir. R.A. van der Eijk	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Summary	The course explains the basic concepts of theoretical and applied soil mechanics and geotechnical engineering. Theoretical soil mechanics includes: Soil Characteristics; Groundwater flow; Geomechanics; consolidation, and Shear Strength of Soils. Applied soil mechanics includes: Retaining Structures; Foundations; and Slope Stability.	
Course Contents	Soil Characteristics: Classification of soils; volumetric and gravimetric relationships for soils. Groundwater: Pore pressure and effective stress; Darcys law, permeability and groundwater flow. Geomechanics: Stresses and initial stress state; strains, stress-strain relationships and tangent modulus; Elastic solutions, drained and undrained behaviour. Consolidation: 1D compression, Oedometer test and data interpretation, consolidation coefficient, settlement Shear Strength of Soils: Drained and undrained soil behaviour; total and effective shear strength parameters; Mohr circles; Mohr-Coulomb failure criterion; direct shear test; triaxial test. Retaining Structures: Lateral earth pressure at rest; passive and active pressures Foundations: Geotechnical bearing capacity, shallow foundations Slope Stability: Limit equilibrium methods.	
Study Goals	The main goal of this course is to: Have an understanding of, and be able to apply the following: Soil classification; Groundwater flow; Stresses in the ground; Strains and stiffness; Consolidation; Shear strength of soils; Laboratory testing; Earth retaining structures; Foundations; Slope stability. By the end of the course the student should be able to: Explain fundamentals of soil behaviour in terms of volume changes and shear strength mobilisation. Explain an engineering strategy for analysing geotechnical problems, involving settlements, consolidation, groundwater flow, basic foundations, retaining structures and slope stability. Analyse appropriately and evaluate geotechnical problems, utilising presented theory, on the following topics: elastic behaviour and settlements, consolidation, groundwater flow, basic foundations, retaining structures and slope stability.	
Education Method	Self-study, recorded lectures, tutorials.	
Literature and Study Materials	Materials - Soil mechanics by A. Verruijt, 2001 - Lecture notes - Recorded videos of the course	
Assessment	Computer exam to test the learning objectives of this course.	
Expected prior Knowledge	Basic mechanics, and Mathematics, knowledge of the concept of stress and strain as well as elasticity.	
Academic Skills	Critical thinking, self-studying, note making, time management, application of theory for problems	
Literature & Study Materials	- Soil mechanics by A. Verruijt, 2001 - Craig's Soil Mechanics by R.F. Craig (and J. Knappet), 2012 - Lecture notes	
Judgement	online Final exam (Brightspace based exam) to test the learning objectives of this course. For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	- Calculator as described in the Examination regulations - Formula sheet	
Collegerama	No	

CT2023	Programmeren in Python	3
Responsible Instructor	Dr. V.L. Markine	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3	
Start Education	1 2 3	
Exam Period	1 2 3	
Course Language	Dutch	
Collegerama	No	

CTB1210	Dynamica & Modelvorming	5
Responsible Instructor	Dr.ir. A. Gavrilidou	
Instructor	Dr. A. Cabboi	
Instructor	Dr. A.A. Núñez Vicencio	
Instructor	Dr.ir. J. Bosboom	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Collegerama	No	

CTB2300	Dynamics of Systems	3
Responsible Instructor	Dr.ir. K.N. van Dalen	
Responsible Instructor	Dr.ir. H. Hendrikse	
Instructor	Dr.ir. H. Hendrikse	
Co-responsible for assignments	Dr.ir. H. Hendrikse	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Dynamics and Modelling Differential Equations for civil engineering	
Parts	The course consists of lectures in which the course content is provided.	
Summary	To be able to model and analyse the dynamic behaviour of several civil engineering systems	
Course Contents	Mechanical system with one degree of freedom, undamped. Formulate equations of motion for free vibration. Forced vibrations for harmonic, exponential, step and impact/pulse loads. Application of initial conditions. Hydraulic systems with one degree of freedom without damping; also free and forced motion. Mechanical system with damping; free and forced vibrations. Hydraulic system with damping; free and forced motion. First order systems as a limit case of 2nd-order systems. Mechanical systems with two degrees of freedom, without damping. Formulate equations of motion (mass matrix, stiffness matrix). Determination of eigenfrequencies and eigenperiods. Forced response for harmonic loads.	
Study Goals	Learning Objectives (LO): LO1: To recognize the properties of structural and hydraulic civil engineering systems (for example mass, elasticity, discharges), that are essential for the modelling of such systems. LO2: To be able to formulate simple equations of motion, using the second Newtons Law, with the help of the displacement method. LO3: To be able to analytically derive solutions for formulated equations of motion. LO4: Interpret solutions with respect to the fundamental dynamics parameters, like: eigenfrequency, eigenperiod, free and forced motion, resonance, and steady-state motion. LO5: To be able to predict design improvements for dynamic systems that have been studied. 30 hours lectures 3 hours exam 51 hours individual study and exam preparation -----+ 84 hours (3 ECTS)	
Education Method	Lectures, tutorials	
Course Relations	Dynamics Structural mechanics courses	
Literature and Study Materials	Study guide with general information (BrightSpace) Lecture Notes: "Dynamica van Systemen", J. Blaauwendraad, versie 2010. Lecture Slides Other course materials: "Dynamica van Systemen, Vragenstukkenbundel (oefen- en tentamenopgaven)", Studentassistenten CT2022, versie 2011.	
Assessment	Written exam, with open questions	
Permitted Materials during Tests	Calculator and a sheet with main formulas. The latter will be given at the exam together with the exam questions.	
Expected prior Knowledge	Dynamica & Modelforming; Constructuemechanica 1,2,3; Differentiaalvergelijkingen.	
Academic Skills	Analytical thinking; Critical appraisal.	
Literature & Study Materials	Study guide with general information (BrightSpace) Lecture Notes: "Dynamica van Systemen", J. Blaauwendraad, versie 2010. Lecture Slides Other course materials: "Dynamica van Systemen, Vragenstukkenbundel (oefen- en tentamenopgaven)", Studentassistenten CT2022, versie 2011.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Calculator and a sheet with main formulas. The latter will be given at the exam together with the exam questions.	
Colleggerama	Yes	

CTB2400	Numerical Mathematics	3
Responsible Instructor	Prof.dr.ir. C. Vuik	
Co-responsible for assignments	Dr.ir. D. den Ouden-van der Horst	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Expected prior knowledge	analysis, calculus, linear algebra, differential equations	
Parts	During the lectures the most important points are highlighted. Motivation for the various parts of the lectures/book are given. Exercises are discussed and explained. There is a lot of material available on the corresponding website for information, motivation, exercises and old exams.	
Summary	After the course the student is able to design a simple numerical method, analyses it and implement it. Furthermore, the student should be able to consider the results of numerical methods, analyse them and explain their behaviour. An oral and/or written report of these findings can be given by the student.	
Education Method	Lecture	
Tags	Mathematics Numeric Methods Programming	
Literature & Study Materials	Numerical Methods for Ordinary Differential Equations C. Vuik, F.J. Vermolen, M.B. van Gijzen and M.J. Vuik TU Delft OPEN, Delft, 2023 https://textbooks.open.tudelft.nl/textbooks/catalog/book/57	
Judgement	Grade of the written test is the final grade. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Only a standard calculator.	
Collegerama	Yes	

IFEEMCS010500	Probability and Statistics	3
Responsible Instructor	Dr. C. Kraaikamp	
Responsible Instructor	Dr. R.J. Fokkink	
Contact Hours / Week x/x/x/x	x/x/4/x or x/x/x/4	
Education Period	3 4	
Start Education	3 4	
Exam Period	3 4 5	
Course Language	Dutch English	
Expected prior knowledge	Linear Algebra, Calculus for Engineering part 1 and part 2 or Calculus for Science part 1 and part 2.	
Course Contents	Graphical and numerical summaries of data Introduction to probability Some elementary combinatorics to find probabilities Conditional probability; Bayes' rule; stochastic (in)dependence Random variables; probability mass function; density function; distribution function Standard distributions: Binomial, Poisson, Geometric, Normal, Uniform, Exponential, Pareto Expectation and variance; transformations; Jensen's inequality Multivariate random variables; joint distributions; joint and marginal density functions; (in)dependence of random variables Covariance and correlation Chebychev's inequality; Law of Large numbers; Central Limit Theorem Sampling theory and statistical models; mean, sample variance, histogram, empirical distribution function, boxplot Theory of Estimators: Bias, Efficiency, Mean squared error Linear Regression: univariate and multivariate, goodness of fit Introduction to statistical learning Confidence intervals of Mean and Proportion; t-distribution Testing theory: Type I/II Error, p-value, significance level, critical region One sample t-test	
Study Goals	At the end of the course, you will be able to: - compute (conditional) probabilities - use discrete and continuous random variables - use joint random variables - compute expectation and (co)variance - apply the law of large numbers and the central limit theorem - analyse data by means of graphical and numerical summaries - formulate a statistical model - estimate parameters - perform a linear regression analysis - construct a confidence interval - formulate and carry out hypothesis tests - Use a computer program to perform elementary simulation and data analysis	
Education Method	Interactive lectures, pre-lecture videos, exercises	
Literature and Study Materials	Book: Mathematical introduction to Probability and Statistics, Dekking et al.	
Assessment	One midterm (week 5) on Part 1: Probability Theory and one endterm (week 10) on Part 2: Statistics. Final grade is the average of both tests. To pass the course, the grade for part 2 needs to be at least 4 or higher.	
Permitted Materials during Tests	1) Formula sheet to be used for exams of Probability and Statistics 2) Any kind of calculator	
Remarks	Please note that this course is available in Q3 and in Q4.	
Co-Instructor	Dr. R.J. Fokkink	
Co-Instructor	Dr. C. Kraaikamp	

IFEEMCS012100	Calculus for Engineering, part 1	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012200	Calculus for Engineering, part 2	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012300	Calculus for Engineering, part 3	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Engineering 2 (IFEEMCS012200) or a similar course.	
Course Contents	Line integrals and surface integrals. Integral theorems of Green, Stokes and Gauss. Sequences and series. Power series and Taylor series.	
Study Goals	After this course the student should be able to 1. find parametrizations of curves and surfaces (in 2D and 3D); 2. calculate a line integral or a surface integral of a function or a vector field; 3. mention the physical interpretation of these line and surface integrals; 4. define and formulate the physical meaning of concepts like gradient, curl and divergence; 5. calculate line and surface integrals via one of the integral theorems (Green, Stokes, Gauss) 6. determine whether a sequence is convergent or divergent, for example by calculating its limit; 7. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 8. determine the interval of convergence of a power series; 9. determine the Taylor series of a simple analytic function, for example $\sin(x)$, $\cos(x)$, $\exp(x)$, $\ln(1+x)$, $\arctan(x)$, $(1+x)^a$.	
Education Method	Lectures (per week 2x2 hours).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital, partly written. See Brightspace for more details. There will be two partial exams, of 90 minutes each: - The first partial exam is about vector calculus; - The second partial exam is about sequences and series. The final grade is the average of the grades for the two partial exams. There is only one resit, about the complete course, that lasts 180 minutes. It is NOT possible to resit only one partial exam.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators allowed.	
Co-Instructor	Dr. F.J. van de Bult	

WI1807TH1-21	Linear Algebra	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programme students are applying for.	
Course Contents	- Systems of linear equations, elementary row-operations, (reduced) echelon form - Vector equations, spans - Solution sets of (non)homogeneous systems of linear equations - Linear independence - Linear transformations and standard matrices - Matrix algebra (sum, product), inverse of a matrix - Subspaces in $\mathbb{R}^n$, column space and null space - Basis and Dimension - Determinants - Eigenvalues and eigenvectors - Diagonalization	
Study Goals	After successfully completing the course you will be able to: - solve systems of linear equations. - identify if a system of linear equations is solvable, and if the solution is unique. - show the equivalence between a vector equation, a matrix equation and a system of linear equations. - give a geometric description of solution sets of vector equations. - determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - identify if a transformation is linear and determine the standard matrix of a linear transformation. - perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - determine the determinant of a matrix by applying the properties of a determinant. - apply the properties of determinants (product, transpose, inverse). - determine the eigenvalues and eigenvectors of a matrix. - determine the algebraic and geometric multiplicity of the eigenvalues of a matrix. - determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - analyze the case of complex eigenvalues and eigenvectors.	
Education Method	Lectures, 4 hours per week (2x2)	
Literature and Study Materials	- Book: David C. Lay, Steven R. Lay, Judi J. McDonald, Linear Algebra and its Applications, Sixth edition, Global Edition, ISBN-13: 978-1292351216. - Slides on Brightspace. - Prelecture videos. - Online homework assignments (Grasple). - Old exams on Brightspace.	
Books	David C. Lay, Steven R. Lay, Judi J. McDonald, Linear Algebra and its Applications, Sixth edition, Global Edition, ISBN-13: 978-1292351216.	
Assessment	Digital exam (in Grasple) on campus.	
Remarks	Lectures will be in Dutch!	
	Exams will be in offered in Dutch and also in English.	
Co-Instructor	Dr. E. Emsiz	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementary Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementary Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Schakelopleiding Civil Engineering

Civil Engineering Fundamentals. Students choose one course out of the following two Civil Engineering Fundamentals courses.

CT1000 S	Structural Mechanics	6
Responsible Instructor	T.R. van Woudenberg	
Instructor	Ir. J.W. Welleman	
Contact Hours / Week x/x/x/x	Class sessions: 3 × 2 hours per week for in total 16 weeks Homework: 4 hours per week for in total 20 weeks	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	see study goals and education method	
Study Goals	At the end of this module, the student has shown it is able to: 1. Model, evaluate and reflect on validity of support reactions, internal forces and displacements in skeletal structures using differential equations. 2. Model, evaluate and reflect on validity of support reactions, internal forces and displacements in statically indeterminate skeletal structures using the force- and hybrid method. 3. Model, evaluate and reflect on validity of the stress state on homogeneous planar elements for any coordinate system. 4. Model, evaluate and reflect on validity of buckling load of skeletal structures. This aligns with the learning objectives of Structural Mechanics 3 of the regular BSc programme of Civil Engineering. As a prerequisite for the main objectives, a students should be able to model, evaluate and reflect on forces, displacements and stresses in statically determinate structures. This is made specific in the following prerequisite learning objectives: a. Model, evaluate and reflect on validity of shape (only for cables), support reactions and internal forces in statically determinate skeletal structures and cables using equilibrium relations and virtual work (only on beam structures), required for learning objective 2 and 2. b. Model, evaluate and reflect on validity of displacements of truss- and skeletal bending structures using Williot, forget-me-nots and Moment-area theorems, required for learning objective 2 and 4 c. Model, evaluate and reflect on validity of normal- and shear stresses in thick- and thin-walled cross-sections due to normal forces, bending moments and torsional moments, required for learning objective 3 The learning objectives a-c align almost entirely with the learning objectives of Structural Mechanics 1 and 2 of the regular BSc programme of Civil Engineering. The learning objectives are summarized in three subjects: - Statically indeterminate structures, including learning objective 1, 2, a and b - Continuum mechanics, including learning objective 4, a and c - Buckling, including learning objective 1, 3 and b	
Education Method	Students start the course with an ungraded, self-assessed progress test to identify shortcomings in their knowledge. In class, student make a personal learning plan on module-level with the results from the self-assessed progress tests. Then, students with similar planned modules in their learning plan are grouped together using the provided literature and exercises. This personal learning plan can be adjusted at a later stage. The learning-line of structural mechanics as taught in the BSc programme of Civil Engineering is offered as separate modules (corresponding to the learning objectives 1-4) in a fixed order (1, 2, 3, 4). In class, each learning objective 1-4 will be introduced classically on a high-level. During this modules, theres enough time to work on the prerequisite modules (corresponding to learning objective a-c). All of the separate modules are collected in an online interactive textbook, breaking down each module into separate parts of theory and skills including exercises. Assistance on learning objective a-c is provided to smaller student groups and individual students. The support by lecturers and TAs in class comprises of: - Classical introduction on learning objective 1-4 - Assistance on making a personal learning plan, if requested - Grouping students by module - Additional explanation to small student groups on the literature, if requested - Guidance and feedback on exercises, if requested	
Course Relations	This course replaces Structural Mechanics 1, 2 and 3 of the regular Bachelor programme as a preparation for the MSc Civil Engineering. It is expecting some prior knowledge on learning objective a c to be able to finish the course on time.	
Assessment	The learning will be assessed incrementally: - Exam period Q1, assignment on statically indeterminate structures - Halfway Q2, assignments on statically indeterminate structures and continuum mechanics - Exam period Q2, assignments on buckling and continuum mechanics - Exam period Q3, assignments on statically indeterminate structures, on buckling and on continuum mechanics	
Expected prior Knowledge	Admission to bridging program is enough. It is expecting some prior knowledge on learning objective a c to be able to finish the course on time.	
Literature & Study Materials	- Engineering Mechanics, Volume 1: Equilibrium, C. Hartsuijker and J.W. Welleman, 2006, Springer - Berlin, ISBN: 978-1-4020-4120-4 - Engineering Mechanics, Volume 2: Stresses, Strains, Displacements, C. Hartsuijker and J.W. Welleman, 2007, Springer - Berlin, ISBN 978-1-4020-4123-5 - Mechanica, deel 3: Statisch onbepaalde constructies en bezwijkanalyse. C. Hartsuijker and J.W. Welleman. 2021, Boom uitgevers Amsterdam, ISBN 9789024438105 (only available in Dutch) - Mechanica, deel 4: Stabiliteit van het evenwicht. C. Hartsuijker and J.W. Welleman. 2023, Boom uitgevers Amsterdam, ISBN 9789024446032 (only available in Dutch) - Introduction to Continuum Mechanics. C. Hartsuijker and J.W. Welleman, 2008, Lecture notes provided via Brightspace - Online interactive textbook combining all sources into modules, breaking down each module into separate parts of theory and skills including exercises. This book links the student to relevant part of the other literature and only provides additional content when the other literature. - Additional exercises in ANS - Self-assessed progress test in ANS - Paper-based exams in ANS	
Judgement	A student passes the courses when he/she scores sufficient (minimum 6.0) for an assignments on each of the subjects. Each subject is graded separately as a partial grade .	
Permitted Materials during Exam	N/A	
Collegerama	No	

CTB2110	Fluid Mechanics	5
Responsible Instructor	Dr.ir. T.S. van den Bremer	
Responsible Instructor	Dr. C. Chassagne	
Instructor	Dr. C. Chassagne	
Instructor	Dr.ir. T.S. van den Bremer	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Collegerama	Yes	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Schakelopleiding Civil Engineering

Civil Engineering Design. Students choose one course out of the following four Civil Engineering Design courses.

CTB3340-15	Building Structures 1	4
Responsible Instructor	Ir. S. Pasterkamp	
Instructor	A.C.B. Schuurman	
Instructor	Dr.ir. G.J.P. Ravenshorst	
Instructor	J. Cupa	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Expected prior knowledge	Ontwerpen van constructies en funderingen Beton en staalconstructies	
Parts	-Workshops special aspects low rise structures -self study of reader and written test -design exercise and presentation	
Course Contents	The course consists of 10 modules: Introduction low rise Low rise steel Low rise in timber Low rise in prefab concrete Low rise in structural glass Floors and foundations Introduction housing Housing in situ Housing in masonry Housing in timber	
Study Goals	Demonstrate understanding of various aspects of low rise buildings and be able to design low rise building structures, with a specific focus on: Functional, structural and construction requirements for housing and industrial buildings Relationship between load bearing structure and finishing structure for roof, facade and ground floor for housing and industrial buildings Steel and concrete elements on intermediate level (know requirements related to transport, assemblage and maintenance, design and calculate of structures with some complexity, select suitable detail and construction options) Timber elements on introductory level (design and calculate simple structures, select suitable detail and construction options, understand thermal behaviour) Glass or masonry elements on introductory level (understand relevant material properties, apply relevant safety procedure, design and calculate simple structures, apply suitable detail options) Stability systems (braced and unbraced structures), with attention to second order forces (understand differences, calculate frameworks with second order effect) Foundations on intermediate level (understand various ways of foundations, be able to calculate stiffness of various foundations) Specific aspects of ground floors for industrial buildings and housing (know various types of ground floors with advantages and disadvantages, apply suitable measures to deal with cracking of concrete ground floors)	
Education Method	Workshops and design exercise	
Course Relations	Building Structures 2	
Literature and Study Materials	Reader Building structures 1, additional study material from workshops to be appointed.	
Assessment	The final grade is made up of a grade for the exam (50%) and a grade for the group assignment (50%). The minimum grade for each part is 5.0.	
Elective	Yes	
Tags	Building Engineering	
Expected prior Knowledge	Ontwerpen van constructies en funderingen. Beton en staalconstructies	
Academic Skills	Integral design approach for various materials	
Literature & Study Materials	Reader building structures 1, other study material provided on Brightspace	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Calculator+quick reference	
Collegerama	No	

CTB3355	Hydraulic Structures	4
Responsible Instructor	Dr.ing. M.Z. Voorendt	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Relations		
Literature and Study Materials		
Permitted Materials during Tests		
Elective	Yes	
Tags	Broad Design Drawing Fluid Mechanics Intensive Soil Mechanics Structural Mechanics Water Engineering	
Judgement		
Collegerama	No	

CTB3370-18	Geometrical Design of Roads and Railways	4
Responsible Instructor	Dr.ir. H. Farah	
Instructor	Prof.dr. Z. Li	
Instructor	Dr.ir. A.M. Salomons	
Contact Hours / Week x/x/x/x	0.0.10.0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Geometric design of roads and railways: Cross sections, horizontal and vertical alignment, integration of the horizontal and vertical alignments; Intersection design, roundabouts and turbo-roundabouts, interchanges, discontinuities; Design for pedestrians and cyclists; Railroad categories, and characteristic of railroad traffic; Functional design of railway-infrastructure; Design of trams in urban areas; Human factors and road systems design; Using AutoCAD and Civil3D;	
Study Goals	At the end of this course you will be able to: Apply design guidelines and design methodologies for designing the geometric alignment of roads and railways. Explain the relationship between road design and road user behaviour. Interpret and analyse characteristics of railway infrastructure elements. Design railways and tram lines. Display roads and railways design using AutoCAD and Civil3D. Evaluate the implications of different design alternatives on road user behaviour, traffic safety and traffic operation. Think critically and make design decisions.	
Education Method	The course Geometric Design of Roads and Railways is taught in lectures and practicum. The lectures explain the theory and principles of roads and railways design, and provide knowledge and insights on the fundamental issues in design. In the practicum (lab hours), students develop skills to apply this knowledge on real projects of designing roads and railways. The practicum is also meant to increase the understanding of the subject and to prepare the students for the assessment.	
Assessment	The assessment in this course is composed of the assessment of the two projects and the written examination (digital examination via ANS) as following: o 35% First project (a minimum score of 5.0 is required) o 25% Second project (a minimum score of 5.0 is required) o 40% Written digital examination with registration on campus (a minimum score of 5.0 is required)	
Expected prior Knowledge	CTB1420 - Transport & Planning	
Academic Skills	- Problem solving - Critical thinking - Design - Reasoning - Writing skills - Presentation skills	
Literature & Study Materials	The following books and guidelines can be referred to: Traffic & Highway Engineering, Fifth Edition, N.J. Garber, L.A. Hoel, 2009, CENGAGE Learning. Designing Safe Road Systems: A Human Factors Perspective (Human Factors in Road and Rail Transport) by Jan Theeuwes, Richard Van Der Horst and Maria Kuiken (Aug 28, 2012). NCHRP Report 600, Human Factors Guidelines for Road Systems, Second Edition, 2012. Behavioural Adaptation and Road Safety: Theory, Evidence and Action by Christina Rudin-Brown and Samantha Jamson (May 24, 2013). Modern Railway Track, Version 3.4, Coenraad Esveld, 2014. ROA, Richtlijn Ontwerp Autosnelwegen 2019, Rijkswaterstaat, 2019 (The Dutch design guidelines for motorways). CROW Handboek Wegontwerp (NL), (available from CROW kennisbank). ASVV-Dutch guidelines in urban areas (NL), (available from CROW kennisbank). CROW, Design manual for bicycle traffic (EN), June 2007. CROW, Road safety manual (EN), May 2009. Besides these references, specific material will also be provided during the lectures.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	No material is allowed during the exam.	
Collegerama	No	

CTB3420	Integral Design of Infrastructure	4
Responsible Instructor	Dr.ir. G.A. van Nederveen	
Instructor	Dr. E.J. Houwing	
Contact Hours / Week x/x/x/x	0/0/0/6 Hoorcollege 2 uur per week Project: groepswork, 4 u/w begeleid	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Parts	- Develop and execute a project plan; - Make a project analysis, including problem analysis, site analysis and stakeholder analysis and definition of the project scope; - Develop a design proposal; go through the full design cycle for a given case; - Realize the integration of the different engineering disciplines; - Elaborate specific parts or aspects; - Oral and written presentation of the design proposal.	
Summary	Integration and application of different civil engineering disciplines in a design project setting.	
Course Contents	Infrastructure projects are characterized by the complex setting in which they take place and the broad spectrum of involved technical disciplines: road, rail, structures, water, ground, etc. In other words: this kind of projects are multi-disciplinary and integral by definition. The planning and design process are executed in groups using a case assignment from practice. Systems engineering is an important method in this course. Subjects: - Project management, - Design approaches, - Structural engineering, - Systems engineering, - Economic and social aspects, sustainability, use and environment. - Risk management.	
Study Goals	- Develop and execute a project plan; - Develop a design proposal; go through the full design cycle for a given case; - Realize the integration of the different engineering disciplines; - Elaborate specific parts or aspects; - Oral and written presentation of the design proposal.	
Education Method	Project course with supporting lectures	
Course Relations	This course builds upon first year course CTB 1220 "Integraal Ontwerp en Beheer" and more generally upon knowledge gained in the first and second year of the Bachelor CT. This course uses and is relevant for all civil engineering disciplines.	
Literature and Study Materials	Will be made available through BrightSpace	
Assessment	Group reports (project plan, project analysis, design documents, final report), group presentations, and individual reports are assessed.	
Expected prior Knowledge	Third year BSc Civil Engineering	
Academic Skills	Collaboration in teams, presenting, reporting, information skills, awareness of societal impact of infrastructure projects	
Literature & Study Materials	Will be made available through Brightspace	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	n/a	
Colleggerama	No	

Dr.ir. J. Bosboom

Unit	Civiele Techniek & Geowetensch
Department	Coastal Engineering
Telephone	+31 15 27 84606
Room	23.HG 3.66

Dr.ir. T.S. van den Bremer

Department	Environmental Fluid Mechanics
Telephone	+31 15 27 83306
Room	23.HG 2.87

Dr. F.J. van de Bult

Unit	Elektrotechn., Wisk. & Inform.
Department	Analysis
Telephone	+31 15 27 82541
Room	36.HB 08.300

Dr. A. Cabboi

Department	Mechanics and Physics of Structures
Telephone	+31 15 27 83419
Room	23.S2 1.54

Dr.ir. N.P.K. Chan

Department	Statistics
Room	36.HB 06.130

Dr. C. Chassagne

Unit	Civiele Techniek & Geowetensch
Department	Environmental Fluid Mechanics
Telephone	+31 15 27 85970
Room	23.KG 03.120

J. Cupa

Department	Applied Mechanics
-------------------	-------------------

Dr.ir. K.N. van Dalen

Unit	Civiele Techniek & Geowetensch
Department	Dynamics of Structures
Telephone	+31 15 27 88388
Room	23.HG 3.61

Ir. R.A. van der Eijk

Department	Geo-engineering
Room	23.KG 00.560

Dr. E. Emsiz

Unit	Elektrotechn., Wisk. & Inform.
Department	Applied Probability
Telephone	+31 15 27 84579
Room	36.HB 07.060

Dr.ir. H. Farah

Unit	Civiele Techniek & Geowetensch
Department	Traffic Systems Engineering
Telephone	+31 15 27 83401
Room	23.HG 4.20

Dr. R.J. Fokkink

Unit	Elektrotechn., Wisk. & Inform.
Department	Applied Probability
Telephone	+31 15 27 89215
Room	28.1.E300

Dr.ir. A. Gavriilidou

Department	Traffic Systems Engineering
Room	23.HG 4.40

Drs. I.A.M. Goddijn

Department	Numerical Analysis
Telephone	+31 15 27 86408

Department	Numerical Analysis
Telephone	+31 15 27 86408

Unit	Elektrotechn., Wisk. & Inform.
Department	Numerical Analysis
Telephone	+31 15 27 86408
Room	36.HB 03.260

Dr.ir. Z.J.G. Gromotka

Department	Mathematical Physics
Telephone	+31 15 27 81887

Dr.ir. H. Hendrikse

Unit	Civiele Techniek & Geowetensch
Department	Offshore Engineering
Telephone	+31 15 27 88223
Room	23.HG 3.71

Prof.dr. M.A. Hicks

Unit	Civiele Techniek & Geowetensch
Department	Geo-engineering
Telephone	+31 15 27 87433
Room	23.KG 00.060

Dr. E.J. Houwing

Department	Integral Design & Management
Room	23.HG 6.71

Dr. C. Kraaikamp

Department	Applied Probability
Telephone	+31 15 27 81910

Unit	Elektrotechn., Wisk. & Inform.
Department	Applied Probability
Telephone	+31 15 27 81910
Room	36.HB 07.300

Prof.dr. Z. Li

Unit	Civiele Techniek & Geowetensch
Department	Railway Engineering
Telephone	+31 15 27 82325
Room	23.S2 2.29

Dr. V.L. Markine

Unit	Civiele Techniek & Geowetensch
Department	Railway Engineering
Telephone	+31 15 27 83206
Room	23.S2 2.33

Dr.ir. G.A. van Nederveen

Unit	Civiele Techniek & Geowetensch
Department	Integral Design & Management
Telephone	+31 15 27 81854
Room	23.HG 6.64

Dr. A.A. Núñez Vicencio

Unit	Civiele Techniek & Geowetensch
Department	Railway Engineering
Telephone	+31 15 27 89355
Room	23.S2 2.35

Dr.ir. D. den Ouden-van der Horst

Unit	Elektrotechn., Wisk. & Inform.
Department	Numerical Analysis
Room	36.HB 03.250

Ir. S. Pasterkamp

Unit	Civiele Techniek & Geowetensch
Department	Applied Mechanics
Telephone	+31 15 27 84982
Room	23.HG 6.46

Dr.ir. G.J.P. Ravenshorst

Unit	Civiele Techniek & Geowetensch
Department	Bio-based Structures & Materials
Telephone	+31 15 27 85721
Room	23.S2 2.57

Dr.ir. A.M. Salomons

Unit	Civiele Techniek & Geowetensch
Department	Traffic Systems Engineering
Telephone	+31 15 27 88556
Room	23.HG 4.45

A.C.B. Schuurman

Department	Applied Mechanics
-------------------	-------------------

Department	Applied Mechanics
Room	23.HG 6.48

Dr.ir. R.F. Swarttouw

Department	Analysis
Telephone	+31 15 27 83634
Room	36.HB 08.290

Dr.ing. M.Z. Voorendt

Unit	Civiele Techniek & Geowetensch
Department	Hydraulic Structures and Flood Risk
Telephone	+31 15 27 83340
Room	23.HG 3.76

Prof.dr.ir. C. Vuik

Unit	Elektrotechn., Wisk. & Inform.
Department	Delft Institute of Applied Mathematics
Telephone	+31 15 27 85530
Room	36.HB 03.320

Ir. J.W. Welleman

Unit	Civiele Techniek & Geowetensch
Department	Applied Mechanics
Telephone	+31 15 27 84856
Room	23.HG 6.55

T.R. van Woudenberg

Department	Applied Mechanics
Telephone	+31 15 27 89739
Room	23.HG 6.45
